The role of persistent organic pollutants in the worldwide epidemic of type 2 diabetes mellitus and the possible connection to Farmed Atlantic Salm...

Rates of type 2 diabetes mellitus (T2DM), both in the United States and worldwide, have been rising at an alarming rate over the last two decades. Because this disease is viewed as primarily being attributable to unhealthy lifestyle habits, a great deal of emphasis has been placed on encouraging increased exercise, better dietary habits, and weight loss. Recent studies reveal that the presence of several persistent organic pollutants (POPs) can confer greater risk for developing the disease than some of the established lifestyle risk factors. In fact, evidence suggests the hypothesis that obesity might only be a significant risk factor when adipose tissue contains high amounts of POPs. Chlorinated pesticides and polychlorinated biphenyls, in particular, have been strongly linked to the development of metabolic syndrome, insulin resistance, and T2DM. In addition to reviewing the evidence associating POPs to these conditions, this article explores the possible contribution of farmed Atlantic salmon - a significant and common dietary source of POPs - with blood sugar dysregulation conditions.